

NIKOLSK, Russian Federation

WMO: 270660

Lat: 59.534N

Lon: 45.461E

Elev: 468

StdP: 14.45

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.7	-18.5	-29.7	1.1	-24.6	-23.3	1.5	-18.5	14.2	19.9	12.4	20.1	1.9	10	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	84.1	67.7	80.5	65.8	77.2	64.1	69.9	80.3	68.1	77.6	66.1	74.0	5.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.2	74.4	64.6	92.7	72.1	62.8	87.1	70.7	34.1	80.5	32.6	78.2	31.1	74.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.6	10.0	DB	-29.2	88.4	6.9	2.9	-34.2	90.5	-38.2	92.1	-42.1	93.8	-47.1	95.9	(4)
(5)			WB	-29.4	72.2	6.9	2.3	-34.4	73.9	-38.4	75.2	-42.2	76.5	-47.2	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	10.5	12.9	24.0	37.8	50.8	59.0	64.1	59.3	49.5	37.5	25.1	16.5
	DBStd	21.00	14.77	13.82	9.67	8.31	9.40	7.82	6.53	7.16	7.13	7.62	10.62	13.96
	HDD50	5841	1224	1038	805	372	109	14	1	7	94	393	747	1037
	HDD65	10210	1689	1458	1270	815	448	212	97	204	465	853	1197	1502
	CDD50	1241	0	0	0	7	133	282	438	297	79	5	0	0
	CDD65	138	0	0	0	0	7	31	70	29	1	0	0	0
	CDH74	1438	0	0	0	1	124	350	680	276	7	0	0	0
	CDH80	347	0	0	0	0	22	81	169	75	0	0	0	0
Wind	WSAvg	4.6	4.7	4.8	5.0	4.7	4.9	4.5	3.8	3.8	4.1	4.9	4.9	5.0
Precipitation	PrecAvg	23.9	1.5	1.1	1.1	1.4	2.0	3.0	3.2	2.6	2.1	2.3	1.9	1.8
	PrecMax	32.2	2.7	2.1	2.5	3.3	5.2	7.0	7.5	5.5	4.8	4.1	4.6	3.4
	PrecMin	18.0	0.3	0.0	0.2	0.4	0.7	0.6	0.9	0.6	0.4	0.0	0.5	0.5
	PrecStd	3.4	0.6	0.5	0.5	0.7	1.1	1.8	1.5	1.2	0.9	0.9	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	36.2	50.0	71.5	83.0	87.9	88.4	87.5	74.7	61.8	47.5	40.1
		MCWB	33.6	34.4	40.3	54.5	63.2	68.5	70.4	70.4	62.0	52.4	45.4	39.4
	2%	DB	33.3	34.2	43.0	63.3	77.3	82.3	84.5	80.7	69.0	55.4	41.9	36.4
		MCWB	32.5	32.9	35.9	49.6	59.8	66.0	68.7	67.1	59.4	51.0	40.3	35.5
	5%	DB	32.1	32.9	39.2	57.6	72.3	78.1	81.3	76.1	64.5	51.6	39.3	33.9
		MCWB	31.3	31.8	34.0	46.1	56.6	63.8	67.0	64.8	57.2	48.1	38.2	32.9
	10%	DB	29.4	30.8	36.4	52.6	67.6	74.4	77.8	71.8	61.2	48.3	36.2	32.4
		MCWB	28.4	29.5	33.0	43.5	54.3	62.2	65.7	62.7	55.4	45.9	35.3	31.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	35.0	41.8	55.4	65.7	70.7	72.9	72.0	63.9	54.9	45.9	39.6
		MCDB	34.3	35.6	48.5	68.5	77.9	82.7	83.3	83.7	71.5	59.0	47.3	40.1
	2%	WB	32.8	33.3	37.6	50.8	62.3	68.3	70.4	68.8	60.7	51.3	40.5	35.7
		MCDB	33.2	34.0	41.0	60.8	72.8	79.4	80.7	78.3	66.5	54.9	41.3	36.4
	5%	WB	31.2	32.0	35.2	47.4	59.1	66.0	68.7	66.1	58.3	48.5	38.0	33.2
		MCDB	32.0	32.9	37.9	55.7	69.5	74.6	78.2	72.9	63.6	51.2	39.0	33.7
	10%	WB	28.5	29.5	33.4	44.1	56.2	63.7	67.1	63.9	55.9	46.0	35.4	31.7
		MCDB	29.4	30.7	35.8	51.5	64.9	71.8	75.5	70.5	60.4	48.4	36.2	32.4
Mean Daily Temperature Range	5% DB	MDBR	11.3	13.4	16.1	17.8	20.2	18.8	18.8	16.8	14.3	8.9	7.1	8.9
		MCDBR	8.6	8.4	19.3	27.6	27.7	25.9	25.0	23.6	21.4	15.6	8.3	6.5
		MCWBR	8.7	7.9	13.3	16.1	12.8	11.3	11.0	10.7	12.5	10.7	7.5	6.6
	5% WB	MCDBR	8.4	7.8	13.5	24.8	23.9	21.7	21.5	20.2	18.1	14.1	7.8	6.8
		MCWBR	8.6	7.6	10.0	15.5	12.3	10.7	10.5	10.4	11.2	10.5	7.5	6.9
Clear-Sky Solar Irradiance	taub		0.265	0.305	0.339	0.380	0.361	0.359	0.393	0.385	0.350	0.329	0.289	0.246
	taud		2.043	2.060	2.055	2.153	2.421	2.449	2.366	2.365	2.446	2.363	2.148	1.996
	Ebn at Noon		158	213	245	258	275	277	263	256	247	211	149	117
	Edh at Noon		16	26	36	40	33	32	34	32	25	20	14	11
All-Sky Solar Radiation	RadAvg		112	327	731	1175	1602	1681	1708	1254	711	300	98	60
	RadStd		19	87	111	140	136	147	135	115	92	48	17	10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	-358	N/A	N/A

Nomenclature: See separate page